

Development of cork–gypsum composites for building applications

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Abstract

This paper presents an experimental analysis on a new composite material, cork–gypsum composite. It is shown that cork and plaster are mutually compatible and that a lot of new building materials can be made by mixing those materials in different volume fractions. Mechanical properties of the cork–gypsum composite have been measured. The acoustic absorption coefficient and thermal conductivity of this new composite have also been experimentally obtained and those values are reported for design purposes. Concerning the acoustical insulation characteristics, this composite is not a sound-absorbing material but a reflecting one, and it needs some kind of perforations to behave as an absorbing construction material for sound and noise. The thermal insulation properties are quite good as a result of the thermal conductivity tests. This new composite material is suggested for use in building applications as partitions. © 1999 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Cork is obtained from the bark of cork tree, known botanically as *Quercus suber*. There is another species of cork-producing oak, *Quercus occidentalis*, described by the Swiss botanist, J. Gay, in 1856. Both species are alike, and differ only in their foliage and in the ripening season of their fruits. The most important one as a cork producer is *Quercus suber*, mainly found in Portu-

gal, Spain and North Africa. Portugal is the world's main cork producer, and Spain is the second one.

The bark of the cork tree is organised of a number of layers denoting the annual formation of the cork in growing rings. Cork trees are striped in summertime at 20–25 years old for the first time, and afterwards every 10–12 years.

Cork is composed of small cells bonded by a natural resin. Each cell is filled with air in excess of CO₂ content. There are approximately ten million cells/cm³ of cork. R. Hooke described the cell geometry as imperfect tetrakaidecahedrons or 14-sided polyhedral (six four-sided and eight six-sided). According to the celebrated Lord Kelvin conjecture, which establishes

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that, a 14-sided surface requires the minimum of material for covering any volume [1]. Thanks to this cellular structure, the cork is compressible, and recovers its original shape after removing stress. Therefore, the cork is a good insulation material.

There are four main building products obtained from natural cork. The first one is natural cork, steam-treated to flatten it and to obtain fine slabs and laminates of very thin dimensions. The second one is agglomerated or expanded cork, dark brown in colour due to the hot process of manufacturing at 350°C and 300 kPa. This material is mainly used as acoustic and thermal insulation. Another well-known product is composition cork, which is made up of smaller granules than agglomerated cork. A binding agent is used instead of the natural resin. The last one is cork-rubber [2].

Some other products obtained from cork have been investigated for building applications, but all of them are founded on those four materials. For instance, some kind of sandwich slabs, plasterboards faced, have been described elsewhere [3], and corkboard baked concrete is been investigated as acoustic absorbing material for highway traffic [4].

Cork is a scarce material. However, although it has the important applications in the construction industry that have been mentioned, there is a lot of cork and cork-wood waste, both from manufacturing industry and from forest cleaning and pruning, without industrial utilisation.

In fact, a great deal of cork granules can be obtained from the pruning of cork tree branches and old trunks. A single Company in Cáceres (Spain), with a single cutting machine, produces 500 tons/year of charcoal from the pruning of cork trees, and just now, they offer via Internet 5000 tons world-wide for business. In this concern, cork granules waste are also used as combustible charcoal material for ceramics, tiles and brick firing processes.

The use of cork granules mixed with some mortar or plaster has not been tested anywhere. However, in our opinion it could be an appropriate material for use in building construction. The authors had also considered the use of cork granules mixed with cement paste. Nevertheless, some exploratory work has shown that there is not compatibility between cement paste and cork granules, meaning that the bonding is bad.

Therefore, this paper collects an experimental analysis on the compatibility between cork granules and plaster, that is to say, hemihydrated gypsum and water, to obtain a new composite material mainly for building applications. The mechanical, thermal and acoustic properties of this new composite material are also reported.

2. Materials

Hemihydrated gypsum is the main component of plaster, a well-known building material, and easily available in a lot of countries. Its chemical formulation is $\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$. It reacts with water in a few minutes, setting and hardening. Hemihydrated gypsum is presented in two different phases, the α -hemihydrate or autoclaved plaster, and the β -hemihydrate or stucco plaster. Commercial plaster for building applications is mainly composed of α -hemihydrate and β -hemihydrate. The water-gypsum ratio in the mixing process can vary from 0.60 to 0.80 or higher. It is difficult to diminish the lower ratio because workability problems arise. For ratios > 0.6 the porosity of the final hydrated product increases as the water-gypsum ratio is higher, lowering the mechanical strength [5].

Plaster is brittle and for some building applications fibre reinforcing is quite necessary, for instance, in slabs for ceiling, partitions and fire protection plasterboards. A lot of different fibres are used by the plasterboard manufacturing industry. Natural and synthetic fibres, impinging better mechanical performance and fire behaviour to plasterboards [6].

The plaster used in this research has been obtained by mixing hemihydrated gypsum E 30 with water, using a water-gypsum ratio of 0.8 by mass. Hemihydrated gypsum E-30 has a nominal content in $\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$ higher than 85%, in β -form. This hydrated product has a nominal flexural strength greater than 3 N/mm² after setting and hardening, as can be experimentally assessed by testing of prismatic specimens $4 \times 4 \times 16$ cm³ in three-point bending tests.

A big cork manufacturers' association, Aplicork (Mérida, Spain) has supplied cork, in form of granules of different sizes and characteristics. These cork granulates are named 'bornizo'; it is obtained as waste of pruning and forest cleaning, and can be mixed with other contaminant products: small branches from overgrowth of weeds, small particles of clays, sands and other minerals, and organic compound of soils. Nevertheless, these contaminant components are always present in minor proportions.

The cork granules have been obtained from the cork bark. The first stripping is realised on the trunk when the cork tree is over 20 years old, and this bark is mainly for grinding purposes. As the trees are growing, the first stripping on branches also produces cork bark for grinding. Sometimes, the second stripping produces cork bark for grinding, as its quality is not high. Besides those granule sources, there is another one from pruning the cork tree branches. The last source to obtain the cork granules is the winter pruning. The season for cork bark stripping is summertime. Nevertheless, a lot

of cork bark is obtained in wintertime because of cleaning of the cork forest. This winter cork is another type of cork granules used in the research.

In this concern, four types of cork bark granules have been used:

WG	Winter cork granules.
VBG	Summer cork granules from branches stripping (Valderrey County)
SBG	Summer cork granules from branches stripping (Sierra Morena County)
PCG	Winter cork granules from pruning and forest cleaning.

The main characteristics of these granular cork samples are as follows:

- Winter cork granular materials (WG and PCG)—WG contains 80% cork granules and 20% sawdust from small branches and overgrowth of weeds. Their density is approximately $0.20\text{--}0.28\text{ g/cm}^3$, and porosity is 85%. Their moisture content is high, between 15 and 30%. PCG contains a lot of small wood fibres and soil contamination: 75% cork granules and 15% wood fibres with mineral additions. Their density is approximately $0.15\text{--}0.20\text{ g/cm}^3$ and porosity is 60%. Their moisture content is higher, between 25 and 30%.
- Summer cork granular materials (VBG and SBG)—both of them have very similar characteristics. They are obtained as first bark of stripped branches. In Portugal, it is named ‘refugo’ bark of the first crop. Their composition is near 100% granular cork without noticeable organic or inorganic contaminants. Their density is approximately $0.100\text{--}0.135\text{ g/cm}^3$ and porosity is 85%. Their moisture content lies between 8 and 10%.

Collected cork was ground in a hammer-mill at 3000

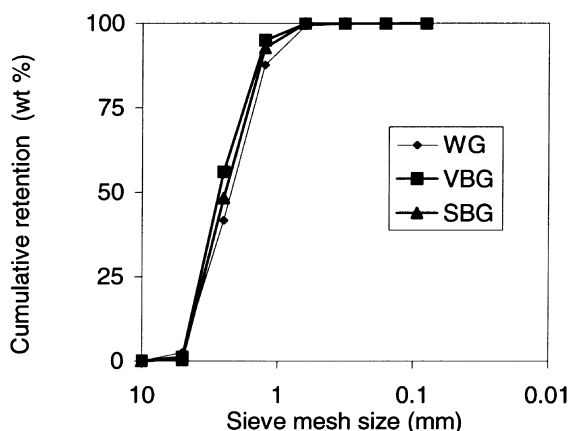


Fig. 1. Cumulative percentage retention curves for three different cork granular materials. Semi-logarithmic scale.

cycles per minute, sieved at sizes $< 12\text{ mm}$, and stored under controlled temperature and humidity conditions in order to obtain moisture content of about 12%, in thermodynamic equilibrium with chamber conditions of 20°C and 65% (RH). Fig. 1 graphically depicts the cumulative percentage grading of three different cork granular materials used in this research.

Chemical composition of cork is well established, and can be obtained elsewhere (See Ref. 1, Chapter IV). Some mechanical properties of cork have been summarised by L.J. Gibson and M.F. Ashby [7].

3. Experiments

3.1. Mechanical properties of cork-plaster composites

3.1.1. Ordinary composites

In order to obtain the mechanical properties of the cork-gypsum composites, several series of prismatic specimens $4 \times 4 \times 16\text{ cm}^3$ have been manufactured. The first group has been realised mixing by mass 80% hemihydrated gypsum with 20% cork granules of each

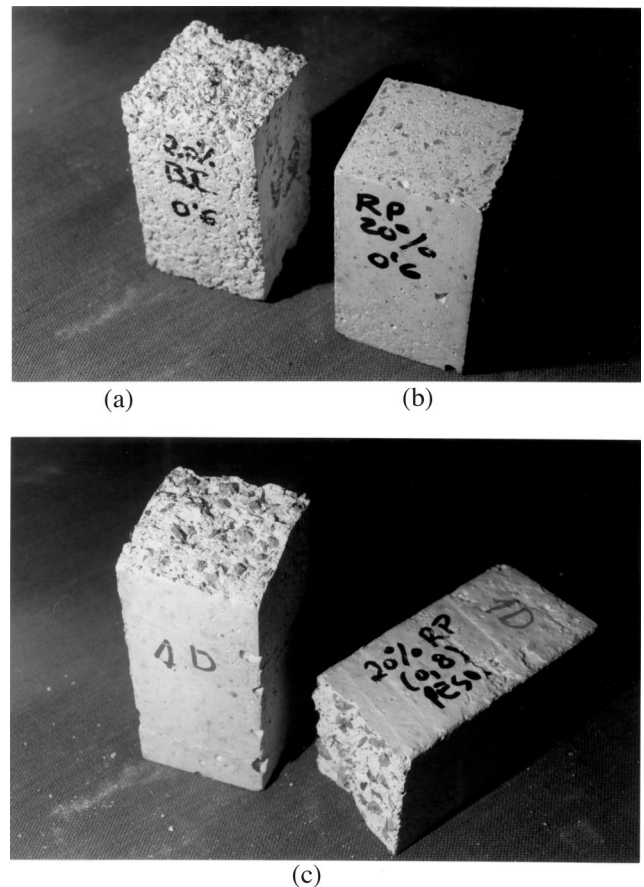


Fig. 2. Cork-gypsum composite specimens after three-point bending test: (a) WG, $w/g = 0.6$; (b) PCG, $w/g = 0.6$; (c) cork-gypsum composite specimens PCG, $w/g = 0.8$, after bending and compression tests.

type. Two different values of water–gypsum ratios (w/g) have been used to analyse its influence on the mechanical properties of the final product: Series 1, w/g = 0.8; and Series 2, w/g = 0.6. Fig. 2 displays the photographs of these specimens.

Specimens have been stored under the controlled humidity conditions (95% (RH)) for a week for setting and hardening. Afterwards, they have been dried to constant mass in a stove at 105°C, in these conditions, bending and compressive tests have been conducted. The results of these experimental tests are shown in Tables 1 and 2.

Workability of Series 1 cork–gypsum composites (w/g = 0.8) is better than that of Series 2 (w/g = 0.6), which is an important factor to take into account. On the other side, the bending and compression tests show similar results for both w/g ratios, in all except those manufactured using PCG cork granules. This cork contains a lot of small vegetable fibres that contribute to increase the mechanical strength of Series 2 cork–gypsum composites (w/g = 0.6) in both bending and compression tests.

Nevertheless, this better mechanical behaviour of PCG cork–gypsum composites, has the counterpart of worst density, because of the higher content on soil waste of this cork granules.

To compare the mechanical performance of ‘bornizo’ with that of PCG as reinforcing materials for plaster, new specimens have been manufactured in similar conditions that Series 1 and 2 above related, but this time using a constant w/g ratio of 0.6. Hemihydrated gypsum of similar characteristics has been mixed with water and with two different fractions of cork granules, that is to say, 10% and 20% by mass, of each class, namely, WG the first one, and PCG the second one, referred to the dry gypsum content.

Table 3 lists the bending and compression tests results, including the density of those two new series of cork–gypsum composites. It is remarkable the lower density for these composites (0.8–1.0 g/cm³ for WG and PCG, respectively) compared with that of similar plasterboard products (density > 1.2 g/cm³).

Table 1

Flexural strength in three-point bending tests of different cork–gypsum composites

Composite series		WG	VBG	SBG	PCG
Series 1 w/g = 0.8	1	2.15	2.10	1.83	2.00
	2	2.30	1.85	1.82	1.90
	3	2.35	1.40	2.19	1.97
	Mean value	2.26	1.97	1.94	1.96
Series 2 w/g = 0.6	4	2.00	2.65	2.68	3.93
	5	2.35	2.65	2.25	3.37
	6	2.50	Null	2.75	3.27
	Mean value	2.28	2.65	2.56	3.52

Table 2

Compressive strength of different cork–gypsum composites

Composite series		WG	VBG	SBG	PCG
Series 1 w/g = 0.8	1	2.26	3.34	3.14	4.14
	2	2.18	3.90	3.64	3.82
	3	2.38	2.87	3.94	3.66
	Mean value	2.27	3.37	3.57	3.87
Series 2 w/g = 0.6	4	2.33	5.12	4.32	6.60
	5	2.38	4.94	3.99	6.51
	6	2.32	Null	4.24	6.64
	Mean value	2.34	5.03	4.18	6.58

3.1.2. Short fibres additions

To analyse the bending strength increase of these new composite materials, small volume fractions of short glass fibres have been added during the mixing process. A 2% by mass of E-glass 2.5 cm long supplied by VETROTEX have been mixed with gypsum–cork (only PCG granules)–water mixtures. The measured fibre diameter is very close to 20 µm, nominal tensile strength: 3400 MPa at room temperature. Young’s modulus: 72 GPa.

Table 4 shows the bending and compression experimental tests results. This small addition of short glass fibres produces quite remarkable increases of both bending and compression strengths.

Such a remarkable increase in mechanical strength may be mainly attributed to the glass fibre strength, and not to the bonding between the hardened gypsum and glass fibres which is poor, as it can be observed in the scanning micrographs (see Fig. 3). On the other hand, it is difficult to avoid clustering phenomena between the glass fibres, which is detrimental to the final mechanical properties of the gypsum–cork–glass fibres composites.

3.1.3. Sandwich-type composites

A lot of sandwich-type specimens have been manufactured for comparison with ordinary specimens of homogeneous structure.

Table 3

Density and mechanical properties of two series of cork–gypsum composites

w/g = 0.6 Composite series		Flexural strength (Mpa)	Compressive strength (Mpa)	Density (g/cm ³)
WG	7	2.17	5.10	0.820
	8	2.07	4.67	0.797
	9	2.21	Null	0.812
	Mean value	2.15	4.88	0.810
PCG	7	3.38	8.17	1.080
	8	3.42	Null	1.068
	Mean value	3.40	8.17	1.074

Table 4
Bending and compressive strengths of PCG cork–gypsum composites reinforced with glass fibres

w/g = 0.6 Sample number	Flexural strength (MPa)	Compressive strength (MPa)
PCG 10	5.95	8.10
PCG 11	8.12	7.77
PCG 12	7.50	Null
Mean value	7.19	7.94

The geometry of sandwich-type specimens is similar to that of ordinary specimens, $4 \times 4 \times 16 \text{ cm}^3$ with a skin thickness of 5 mm. The core has been manufactured by mixing gypsum (w/g = 0.8) with PCG cork granules (20% by mass) and glass fibres (2% by mass). The skin contains the gypsum mixed with glass fibres in the same composition as the core material, but excluding the cork granulates.

The layout is quite easy, and can be automated to obtain large panels of this sandwich-type composite, by lamination or moulding procedures. The mechanical strength of these new products is close to the above referred: 0.5 MPa bending stress, and 0.7 MPa compressive stress.

3.2. Microstructure of gypsum–cork composites

The main characteristics of the microstructures of cork–gypsum composites have been observed by means of scanning microscopy, as represented in Fig. 4.

Samples of cork–gypsum composites after setting and hardening have been analysed to obtain information about the interface compatibility between gypsum and cork. As it can be seen at the micrographs, there exists a perfect bonding between hydrated gypsum crystals and cork cells.

The particular cork cells microstructure can be observed, and gypsum crystals penetrate the open cells at the interface, in such a way that the bonding can be attributed to the mechanical contact without apparent chemical reaction. The pH of gypsum (close to 6.5) contributes to the stability and durability of this bonding with the natural cork cells.

3.3. Acoustic properties of cork–gypsum composites

The acoustic absorption coefficient and acoustic impedance of cork–gypsum composites have been experimentally obtained by means of a Kundt apparatus (Kundt tube), model 4002, equipped with two tubes. The first one having a 3-cm diameter can be used for high sound frequencies (from 800 to 6500 Hz). The second one, having a 10-cm diameter is designed for low frequencies (from 90 to 1800 Hz). The Kundt apparatus is equipped with a sine wave generator mod. 1023, a frequencies analyser mod. 2120 and a Heterodino filter mod. 2020 for direct reading of absorption coefficient of the specimens tested.

Disk-shaped specimens have been manufactured by mixing 20% WG cork and 80% hemihydrated gypsum by mass at 0.8 w/g ratio. Two kinds of specimens of 3 and 10 cm in diameter and having 1.5 and 7.5 cm in thickness respectively have been tested for high and low sound frequencies respectively.

Figs. 5 and 6 illustrate the absorption coefficient of the cork–gypsum composites experimentally measured for low and high frequencies.

Keeping in mind the well known shortcomings of this experimental set-up, it is evident that the cork–gypsum composites have a low absorption coefficient for acoustic waves, mainly in the low-frequency range, what would be discouraging by comparison with the high acoustic absorption coefficient of commercial cork products.

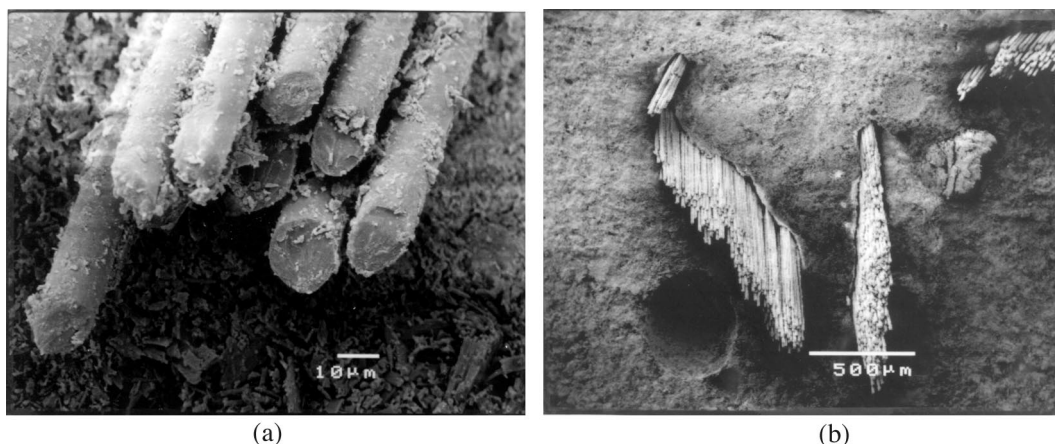


Fig. 3. Scanning micrographs of glass fibres in a plaster matrix. (a) Fibres–gypsum crystals interaction. (b) Bundle of glass fibres.

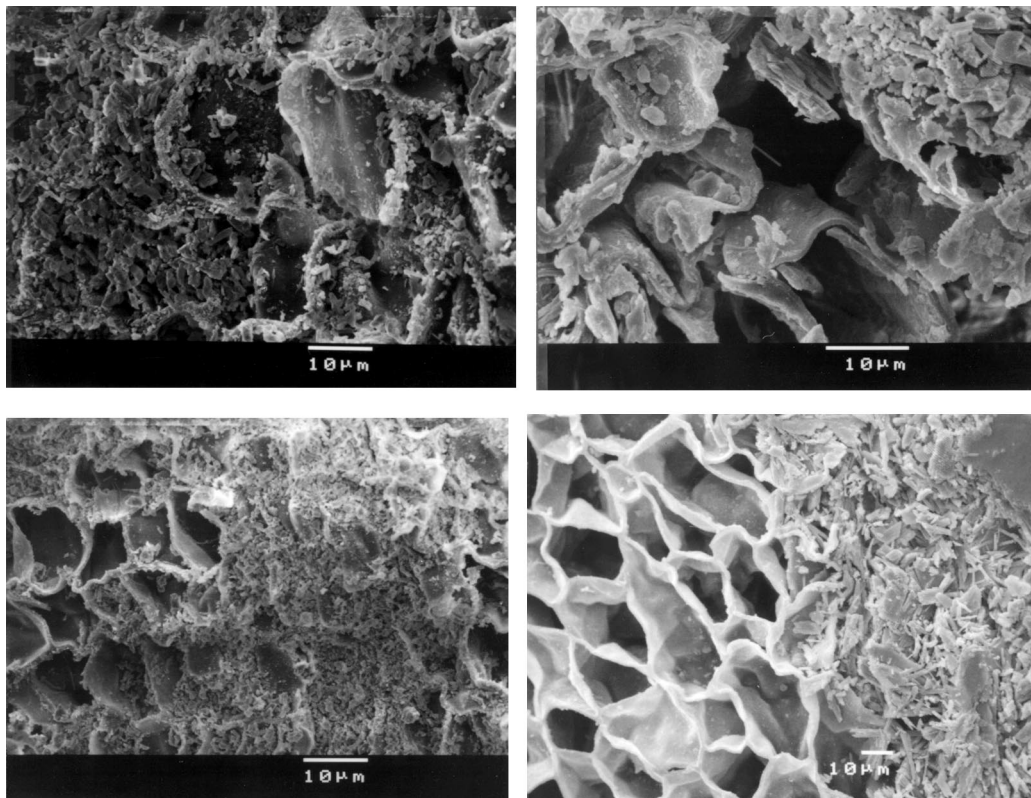


Fig. 4. Four different scanning micrographs showing the details of the interface between cork granules and gypsum matrix.

In fact, adopting for comparison the NRC (Noise Reduction Coefficient), that is to say, the arithmetic mean value of absorption coefficients at 250, 500, 1000 and 2000 Hz, the absorption coefficients of the cork–gypsum composites behave between 0.2 and 0.3 as measured in a Kundt tube. On the other side, similar measurements on agglomerated corkboards give mean values between 0.5 and 0.7 on the same apparatus.

Nevertheless, this is an expected result, if the high acoustic reflection coefficient of gypsum, which render it a good insulation material for air sonic noise is taken into account. On the other side, it is well known that the open porosity of plaster is quite dependent on w/g ratio in mixing and setting. For w/g ratios from 0.6 to 0.8, the open porosity of plaster is between 0.2 and 0.4, which gives a NRC value for plaster between 0.2 and 0.5 for normal incidence.

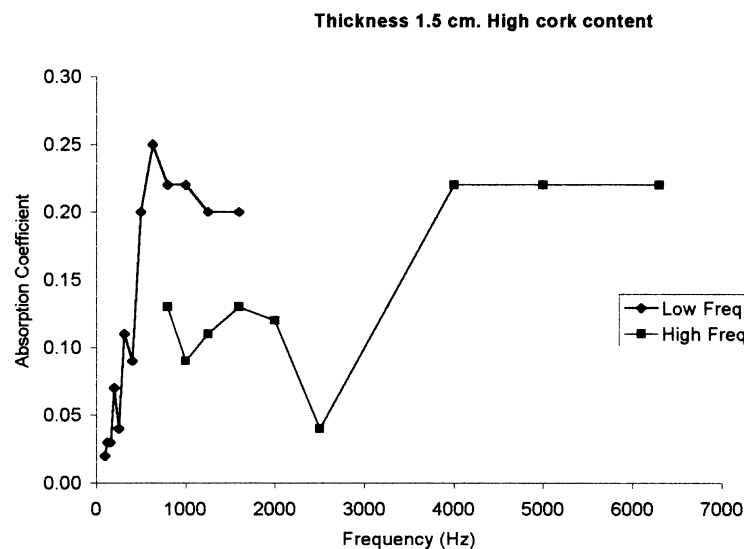


Fig. 5. Absorption coefficient of cork–gypsum composites.

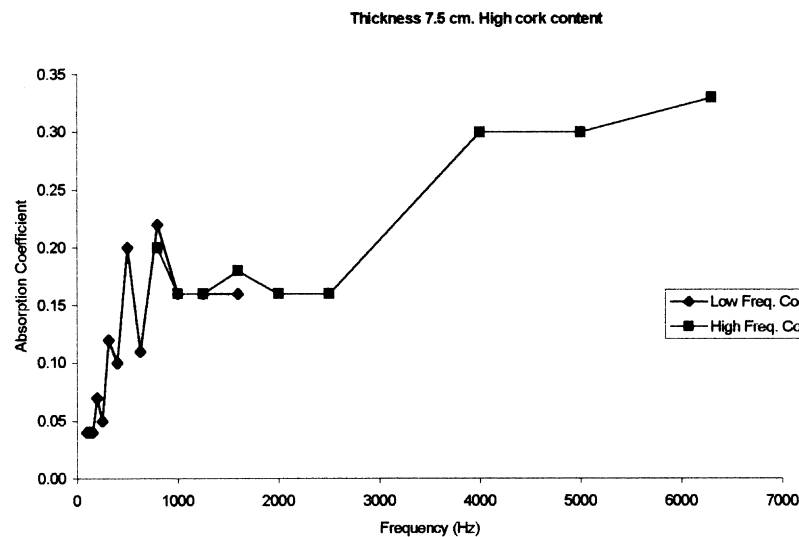


Fig. 6. Absorption coefficient of cork–gypsum composites.

Besides above arguments, it may be argued that during the setting of the gypsum the expansive hydration products are elastically restrained by the cork granules, which reduce the plaster open porosity and explain the low absorption coefficient for the cork–gypsum composites. This can justify the similar results obtained for the specimens of 1.5 and 7.5 cm in thickness.

3.4. Thermal conductivity of cork–gypsum composites

The thermal conductivity of cork–gypsum composites has been measured by means of two different apparatus, obtaining similar results.

The first facility is the Anacon TCA12 (Thermal Conductivity Analyser), the second is the Kemtherm QTM-D3.

Specimens manufactured with similar compositions just described have been tested on these two different apparatus. That is to say, by mixing WG cork, hemihydrated gypsum and water at 0.8 w/g ratio in such proportions to obtain three different densities. The thermal conductivity of the cork–gypsum composites is shown in Table 5, where the density of the specimens can be seen.

There is a correlation between the density and thermal conductivity. Moreover, these measured values are

close to that of building materials used for thermal insulation, which suggests the utilisation of cork–gypsum composites as alternative material for thermal insulation purposes.

In fact, some nominal values of the thermal conductivity (W/m K) are as follows:

Gypsum–mineral perlite composite	0.18
Plasterboard	0.30
Lightweight concrete	0.33
Lightweight plasterboard	0.18
Agglomerated wood particleboard	0.08

On the other side, the density is almost always better for the cork–gypsum composites.

4. Conclusions

The compatibility between cork granules and hemihydrated gypsum has been experimentally tested and examined by means of scanning microscopy. There is a good interaction between the gypsum matrix and cork granules. Nevertheless, it appears that the mechanical properties of this new composite are poor, as a general rule, and other reinforcing agents—glass fibres, for instance—are necessary to increase the bending and compressive strengths of these composites.

Its low density suggests the manufacturing of partitions for interior building compartments, and the cork granules can be good substitution materials for mineral low density additions to gypsum boards and sprayed lightweight plaster.

The acoustic absorption coefficient and thermal conductivity of this new composite have also been exper-

Table 5
Thermal conductivity of cork–gypsum composite

Type of composite	Density (kg/m ³)	k (W/m K) ANACON	k (W/m K) KEMTHERM	k (W/m K) Mean value
1	578	0.1255	—	0.1255
2	600	0.1803	0.1935	0.1869
3	864	0.1995	—	0.1995

imentally obtained and those values are reported for design purposes.

Concerning the acoustical insulation characteristics, this composite is not a sound-absorbing material but a reflecting one. It is similar to plasterboards plates for ceiling and wall partitions, and it needs some kind of perforations to behave an absorbing construction material for sound and noise.

The thermal insulation properties are quite good as a result of the thermal conductivity tests.

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